

(c) Natural Frequency:

EOM: *Cartesian model*

$$m\ddot{x} + kx = 0$$

We know the solution form: $x = X \sin(\omega_n t + \phi)$

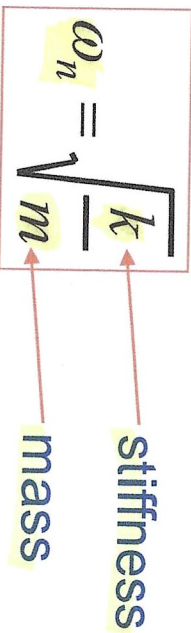
Substituting in EOM, $\ddot{x} = -\omega_n^2 X \sin(\omega_n t + \phi)$

$$-m.\omega_n^2 \cancel{X \sin(\omega_n t + \phi)} + k.\cancel{X \sin(\omega_n t + \phi)} = 0$$

$$-m.\omega_n^2 + k = 0 \Rightarrow \omega_n = \sqrt{k/m}$$

Natural frequency (rad/s):
(ω = angular velocity)

$$\omega_n = \sqrt{\frac{k}{m}}$$



(21)

Natural frequency (Hz):

$$f_n = \frac{\omega_n}{2\pi}$$

(22)

Natural period (s):

$$\tau_n = \frac{1}{f_n}$$

(23)

Tutorial 8
(Lecture Notes #2)

①

(b) Natural Frequency:

EOM: *Polar model*

$$J_o \ddot{\theta} + k_\theta \theta = 0$$

We know the solution form: $\theta = \Theta \sin(\omega_n t + \phi)$

Substituting in EOM, $\ddot{\theta} = -\omega_n^2 \Theta \sin(\omega_n t + \phi)$

$$-J_o \omega_n^2 \cancel{\Theta} \sin(\omega_n t + \phi) + k_\theta \cancel{\Theta} \sin(\omega_n t + \phi) = 0$$

$$-J_o \omega_n^2 + k_\theta = 0 \Rightarrow \omega_n = \sqrt{k_\theta / J_o}$$

Natural frequency (rad/s): ω_n
(ω = angular velocity)

$$\omega_n = \sqrt{\frac{k_\theta}{J_o}}$$

Torsional stiffness
of the shaft
Polar mass moment
of inertia of the disc

(38)

Natural frequency (Hz):

$$f_n = \frac{\omega_n}{2\pi}$$

(39)

Natural period (s):

$$\tau_n = \frac{1}{f_n}$$

(40)

(c) Free Vibration Response: Similar to that of the pendulum

2.5 A Summary of Eqns so far

(3)

	Angular / Rotational Vibrating Systems	Rectilinear Vibrating Systems
EOM	$(\dots)\ddot{\theta} + (\dots)\theta = 0$ J_o → Mass moment of inertia about the centre of rotation (kg.m²) k_θ → Rotational stiffness (Nm/rad)	$(\dots)\ddot{x} + (\dots)x = 0$ m → Mass (kg) k → Stiffness (N/m)
Solution	$\theta = \Theta \sin(\omega_n t + \phi)$	$x = X \sin(\omega_n t + \phi)$
Natural Frequency	$\omega_n = \sqrt{\frac{k_\theta}{J_o}}$	$\omega_n = \sqrt{\frac{k}{m}}$
Initial Conditions	$\theta(0) = \theta_0$ $\dot{\theta}(0) = \dot{\theta}_0$	$x(0) = x_0$ $\dot{x}(0) = \dot{x}_0$

Summary

4

- We have seen that the general solution to the ODE

$$L\ddot{\theta} + g\theta = 0 \quad (1)$$

can be expressed in any of the **three equivalent forms**:

$$\theta = C_1 e^{+i\omega_n t} + C_2 e^{-i\omega_n t} \quad (8)$$

$$\omega_n = \sqrt{g/L}$$

$$\theta = A \cos \omega_n t + B \sin \omega_n t \quad (12)$$

$$\Theta = \sqrt{A^2 + B^2}$$

$$\theta = \Theta \sin(\omega_n t + \phi) \quad (13)$$

$$\phi = \tan^{-1} \left(\frac{A}{B} \right)$$

- The last two forms, viz., Eq. (12) and (13), will be frequently used in this course.

Appendix 2.5

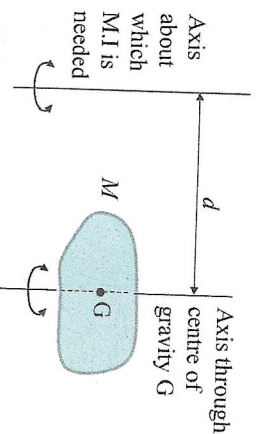
A List of Mass Moment of Inertia Formulas

The mass moments on inertia formulas listed in this section will be given in the exams, if needed.

However, the formulas already shown/used in lecture slides, lecture examples and tutorials will not be given in exams. So, you must note them down in your reference (cheat) sheet.

53

Parallel axis theorem (of moment of inertia of a solid body)



$$J = J_G + M d^2$$

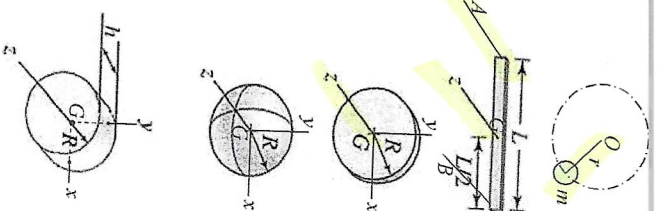
$J \rightarrow$ M.I needed

$J_G \rightarrow$ M.I about its own centre of gravity

$M \rightarrow$ Mass of the body

$d \rightarrow$ Distance between axis

55



$J_o = mr^2$ (for point mass)
(already shown/used in lecture slides)

$J_z = \frac{1}{12} mL^2$ $J_A = J_B = \frac{1}{3} mL^2$
(already shown/used in tutorial solution)

$J_z = \frac{1}{2} mR^2$ $J_x = J_y = \frac{1}{4} mR^2$
(already shown/used in lecture slides)

$J_z = \frac{2}{5} mR^2$ $J_x = J_y = J_z$

$J_x = J_y = \frac{1}{12} m(3R^2 + h^2)$

$J_z = \frac{1}{2} mR^2$

54

End of Chapter 2

(Undamped Free Vibration of 1-DOF Systems)

56